

Narrative Shift Detection in News Media Using Temporal Contrastive Learning

github.com/meet6868/Narrative-Shift-Detection

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Abstract

Narrative shift detection in news media requires tracking how topic framing changes over time, beyond simple similarity or static clustering. This paper presents a generalized, topic-agnostic framework based on Temporal Contrastive Learning (TCL) that detects narrative shifts without supervised narrative labels or topic-specific retraining. We build and analyze the system on 1.17 million articles from eight heterogeneous sources, which are cleaned into 355,334 English articles and segmented into 3.08 million sentences, with 1.34 million topic-classified sentences retained after topic-aware filtering across War, Health, Technology, Climate, and Economics. The framework evolves through five approaches, moving from fixed-window similarity modeling to adaptive segmentation, topic separation, and entity-aware learning. Experimental results show that simple similarity-based methods are insufficient for capturing narrative evolution, while TCL substantially improves temporal consistency and topic coherence. Approach 4 achieves the strongest balance of temporal structure and semantic coherence, and Approach 5 obtains the best overall performance but remains sensitive to multi-loss optimization. Compared with calibrated SBERT drift detection and K-Means clustering, the proposed framework provides stronger narrative modeling and sentence-level interpretability, highlighting the value of temporal, topic, and entity-aware representation learning for topic aware narrative shift.

Keywords: Narrative shift detection, temporal contrastive learning, news media analysis, semantic change detection, interpretable NLP

1 Introduction

Narratives in news media are not static. As events unfold, political contexts change, and public sentiment shifts, the framing, emphasis, tone, and interpretation of facts evolve over time. Consider

the COVID-19 pandemic: initial media coverage framed it as a “foreign virus” in early 2020, shifted to a “global health crisis” by mid-2020, and further transformed to focus on “vaccine rollout and economic recovery” by 2021. While multiple articles may discuss the same topic and entity, their narrative framing often changes gradually or abruptly, reflecting shifts in public discourse, policy changes, or evolving understanding of events.

1.1 Motivation

Raising Awareness About Narrative Shifts in News Media: News media houses continuously build and reshape narratives around topics over time, often reflecting bias patterns and shifting coverage priorities. By understanding and detecting these narrative shifts, this transparency is crucial for media analysis, discourse tracking, and misinformation monitoring. Citizens and researchers need tools to understand how news narratives evolve and what linguistic markers indicate these shifts.

Building a Topic-Agnostic Framework: Existing narrative-shift research is mostly limited to specific topics and entities, with models trained for single events (such as the Ukraine-Russia war or COVID-19) that do not transfer well to new domains, making separate model-building for every topic-entity pair expensive and unscalable. To overcome this, we aim to develop one generalized, topic-agnostic framework that detects narrative shifts without topic-specific retraining, and we evaluate it across five diverse domains: war, climate, health, economics, and technology. Our approach is built iteratively, starting from simple formulations and progressively improving them by addressing observed weaknesses through adaptive segmentation, topic-aware constraints, and entity-aware robustness, showing that a single model can do narrative-shift detection for real-world media analysis which contain diverse topics.

1.2 Problem Statement

Given a sequence of time-indexed news articles, the goal is to automatically detect when a narrative shift occurs according to the topic user gave. Model how the narrative changes over time, identify the specific sentences responsible for the shift, all without requiring supervised narrative labels or topic-specific training.

1.3 Objectives

1. Build a topic-agnostic model that detects narrative shifts across multiple unrelated topics (War, Climate, Health, Economics, Technology) without topic-specific retraining or fine-tuning.
2. Identify and extract the specific sentences that drive narrative shifts, enabling users to understand what changed.
3. Capture both gradual and abrupt narrative changes through adaptive temporal segmentation that respects topic-specific narrative rhythms.

2 Literature Study

Prior research on semantic and narrative change is dominated by lexical semantic shift and contextualized change detection. Diachronic word embeddings reveal that meaning shifts can be captured through distributional changes over time, but the analysis remains word-centric. A survey of diachronic embeddings organizes methods and challenges for semantic shift detection, reinforcing that most approaches still operate at the lexical level. Contextualized semantic shift detection moves to contextual embeddings and proposes a framework of meaning representation, time-awareness, and learning modality, while highlighting unresolved issues in scalability and interpretability. Recent narrative framing and media discourse studies add event-focused signals for better understanding of narrative shifts in complex domains. These works collectively motivate a narrative-level, topic-aware method that explains shifts at the sentence level.

Temporal shift effects are also evident in time-aware modeling and drift analysis. Studies on temporal effects in language models show that data drift over time degrades performance and that semantic drift can be quantified with embedding-based measures, including SBERT. These foundational concepts inform our temporal modeling approach.

Contrastive learning provides a strong representation learning backbone for temporal semantics. SimCLR (Chen et al., 2020) establishes instance discrimination as a scalable self-supervised objective. Foundational metric-learning objectives such as contrastive predictive coding (Chen et al., 2020) show how similarity-driven embeddings can structure representation spaces. Temporal contrastive learning in time-series domains enforces temporal distinctiveness. These insights directly inform our Temporal Contrastive Learning approach for narrative shifts, where temporally adjacent windows are pulled together while narrative changes are pushed apart.

3 Dataset Details

3.1 Data Collection

We collected news articles from 8 heterogeneous sources to ensure diversity. The dataset spans 2011–2025, enabling robust longitudinal narrative shift analysis. The news covers major key domains such as war, economics, health, climate, and technology. Table 1 summarizes the sources, counts, and focus areas.

Source	Count	Format	Focus
NewsSumm Dataset (MDPI)	~50K	JSON	Human-annotated multi-document summaries
CNN Articles (Kaggle)	~80K	CSV	2011-2022 cleaned articles
Global News (Kaggle)	~600K	CSV	Multiple sources, global coverage
Webhose (GitHub)	~200K	JSON	Disaster/accident news
NewsData.io API	~200K	CSV	Structured metadata
Total	1,167,047	Mixed	14+ years temporal coverage

Table 1: Data collection from 8 heterogeneous sources spanning 2011–2025.

3.2 Ideal Reference Articles for Topic Identification

To support topic-aware labeling and filtering, we curated 35 ideal reference articles, roughly seven per topic, from authoritative sources such as Reuters, WHO, NCBI, MIT Technology Review, OECD, IEA, IPCC, IMF, World Bank, and related institutions. These articles act as stable topic prototypes, helping define the semantic space of each topic and enabling centroid-based filtering for War, Health, Technology, Climate, and Economics. Because they capture timeless topic-defining concepts

rather than event-specific language, they do not need frequent updating and provide a consistent reference point for downstream analysis.

Curation Methodology: Table 2 outlines the curation criteria and coverage.

Aspect	Details
Total Articles	35 articles (~7 per topic)
Topics	Cov- War, Health, Technology, Climate, Economics
Selection Process	Manual review and curation from trusted authoritative sources
Extraction Method	Carefully identified and extracted sentences containing topic-defining keywords
Curator Expertise	Domain-aware selection ensuring representative coverage

Table 2: Curation methodology for ideal reference articles.

4 Methodology

4.1 Data Preprocessing Pipeline

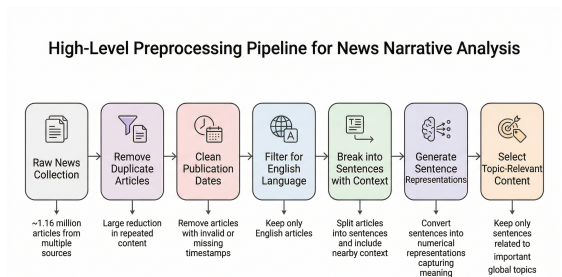


Figure 1: High-Level Preprocessing Pipeline for News Narrative Analysis. The pipeline transforms large-scale, noisy news data into clean, structured, and topic-focused sentence representations.

The pipeline transforms large-scale, noisy news data into clean, structured, and topic-focused sentence representations for narrative analysis. It begins with multi-source article collection, followed by aggressive deduplication to eliminate repeated reporting across outlets. Temporal consistency is ensured through strict date validation, preserving reliable chronological order for downstream shift detection. Language filtering retains only English content to maintain modeling consistency. Articles are then segmented into sentences with surrounding context to preserve narrative flow, and these contextual units are converted into dense semantic representations. Finally, topic-based filtering retains only domain-relevant content, improving signal quality for temporal modeling.

Key Statistics:

- **Preprocessing:** Cleaning steps (deduplication, date validation, language filtering) reduce noise by 69.56%, resulting in 355,334 high-quality articles.
- **Sentence-Level Processing:** Articles are converted into 3,080,512 context-aware sentences (~8.7 per article), with 1,335,158 retained after topic filtering.
- **Topic Focus:** Approximately 43.36% of sentences are preserved, aligned with key domains such as war, economics, health, climate, and technology.

Final Topic Distribution: Table 3 reports the sentence counts and shares after topic filtering.

Topic	Sentences	Share
War	489K	36.7%
Economics	277K	20.8%
Technology	191K	14.3%
Health	188K	14.1%
Climate	188K	14.1%

Table 3: Final topic distribution after preprocessing and filtering.

4.2 Experimental Details

In this project, we use Temporal Contrastive Learning (TCL) to detect narrative shift and compare it against baseline models such as temporal shift detection with SBERT and semantic drift using K-Means clustering. The project also examines how the TCL model evolves over time as it encounters different narrative challenges.

4.2.1 Temporal Shift with SBERT

Background: Sentence-BERT (SBERT) (Reimers and Gurevych, 2019) is a pre-trained transformer model that produces dense, fixed-dimensional sentence embeddings optimized for semantic similarity tasks. The all-mpnet-base-v2 variant generates 768-dimensional representations that capture semantic relationships between sentences.

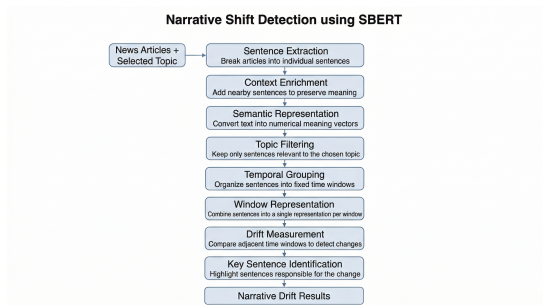


Figure 2: SBERT Model Pipeline for Semantic Representation and temporal drift detection baseline.

Semantic drift refers to changes in the distributional properties of embeddings over time—when the semantic space of topic-related sentences shifts, the centroids and spread of embeddings change, indicating a narrative shift.

SBERT Semantic Drift Baseline:

- 8-stage runtime pipeline in `SBERT_semantic_drift/detect_drift.py`
- SBERT model: `all-mpnet-base-v2` with context variants `w1`, `w3`, `w5` (half-width 0/1/2)
- Fixed temporal grouping into 5-day bins (`WINDOW_DAYS = 5`)
- Batch size: 32, minimum sentence length: 20, top drift-driving sentences: 5
- Topic relevance filtering uses per-topic thresholds from `topic_thresholds.json`
- Drift detection uses per-topic/per-context thresholds computed as `mean + std`

Why Compare: SBERT serves as a strong, well-established baseline because:

1. **Industry-standard embeddings:** Pre-trained on 215M+ sentence pairs, requiring no additional training
2. **Unsupervised drift detection:** Can detect semantic changes without labeled shift annotations
3. **Single-model approach:** Uses static embeddings without temporal contrastive learning, allowing isolation of TCL benefits
4. **Practical deployment:** SBERT is widely used in production NLP systems, making comparison meaningful
5. **Simplicity:** Minimal hyperparameter tuning, providing a clean baseline

4.2.2 Semantic Drift using K-Means Clustering

Background: The K-Means (MacQueen, 1967) baseline groups sentence embeddings into cluster centroids and tracks distributional changes across time windows.

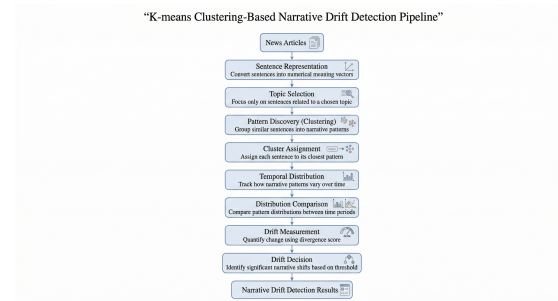


Figure 3: K-Means Pipeline for Drift Detection using cluster distribution shifts.

After embedding generation, samples are assigned to clusters and compared between consecutive periods using divergence-based metrics, which provides an unsupervised reference signal for narrative shift detection.

K-Means Clustering Baseline:

- $K=5$ clusters on topic embeddings
- Jensen-Shannon divergence for distribution shift
- Trained on representative sample (Climate.csv, 1.26GB)
- Drift threshold: 0.3

Why Compare: K-Means clustering provides an alternative baseline that:

1. **Respects topic structure:** Explicitly groups similar days into $K=5$ clusters, one interpretable cluster per topic
2. **Distribution-based detection:** Captures shifts in how days are distributed across clusters, not just point-wise changes
3. **Classical method:** Represents traditional unsupervised approaches, allowing assessment of gains from modern contrastive learning
4. **Simplicity:** No neural networks, making it computationally lightweight and easier to debug
5. **Interpretability:** Cluster centroids can be visualized

4.2.3 Temporal Contrastive Learning (TCL)

Temporal Contrastive Learning is a representation-learning framework for time-ordered text streams. In TCL, each temporal segment (fixed window or adaptive segment) is encoded into an embedding, and training is performed by contrasting pairs of segments: temporally adjacent segments are treated as positive pairs, while non-adjacent or cross-topic segments are treated as negatives. The objective is to learn a space where normal temporal continuity stays close and true narrative changes become separable.

Technical details of TCL approach:

- **Input representation:** Daily/segment-level vectors built from SBERT sentence embeddings (768-D), optionally augmented with topic/entity features.
- **Positive pair:** (z_t, z_{t+1}) from consecutive windows.
- **Negative pairs:** (z_t, z_j) where $j \neq t + 1$ (random, hard, or cross-topic negatives).
- **Encoder:** Transformer-based temporal encoder maps sequence windows to normalized embeddings.
- **Core loss / Temporal Loss (contrastive):** (Oord et al., 2018; Chen et al., 2020)

$$\mathcal{L}_{\text{NT-Xent}} = -\log \frac{\exp(\text{sim}(z_i, z_i^+)/\tau)}{\sum_k \exp(\text{sim}(z_i, z_k)/\tau)} \quad (1)$$

where $\text{sim}(\cdot)$ is cosine similarity and τ is temperature.

- **Shift inference signal:** Drift score from cosine distance between consecutive window embeddings; high distance indicates potential narrative shift.

Why TCL is relevant in our project:

- Narrative shift detection is fundamentally temporal; TCL explicitly optimizes temporal behavior rather than only static similarity.
- It works without manual shift labels, which is practical for large-scale news corpora.
- It improves both detection quality and interpretability when combined with adaptive segmentation and sentence-level evidence extraction.

How our TCL approach evolved over time:

- **Approach 1:** Fixed day windows (overlap/no-overlap) with InfoNCE baseline.
- **Approach 2:** Group-based temporal windows (count/date variants) with stronger NT-Xent setup.
- **Approach 3:** Topic-specific adaptive windows proposed as research direction (not fully implemented).
- **Approach 4:** Ruptures-based adaptive segmentation + multi-objective TCL (current best).
- **Approach 5:** Entity-aware TCL with NER-enhanced features for production-oriented robustness.

Table 4 summarizes the evolution across approaches and their key metrics.

4.3 TCL Approach Evolution

4.3.1 Approach 1: Baseline Day-Level Windowing

Why We Used This Approach:

- Prior baselines (SBERT drift, K-Means clustering) ignore temporal structure and treat points independently.
- Sentence-level modeling misses multi-sentence context and smooth transitions across time.
- Narrative evolution is a temporal process, not a set of independent points.

Inspiration: This approach is mainly inspired by contrastive learning and its temporal extensions. From SimCLR (Chen et al., 2020), we take the idea of learning by pulling similar samples closer and pushing dissimilar samples apart. From TS-TCC (Eldele et al., 2021), we take temporal consistency (nearby time = similar) and context-based learning (use window instead of single point).

Approach	Key Innovation	Loss Function	Inter-Topic	Intra-Topic	Status
1 (No Overlap)	Fixed window W=3, S=3	InfoNCE	-0.0457	0.2182	Completed
1 (Overlap)	Fixed window W=3, S=1	InfoNCE	-0.0286	0.1431	Completed
2 (Count)	4-day grouping	NT-Xent + Hard	0.0114	0.3365	Completed
2 (Date)	5-day windows	NT-Xent + Hard	-0.0361	0.3185	Completed
3	Topic-wise windows	N/A	N/A	N/A	Research
4	Ruptures segmentation	Multi-objective	-0.0875	0.9997	Completed
5	NER + Entity-aware	Multi + Entity	TBD	TBD	To be tuned

Table 4: Approach comparison and evolution summary.

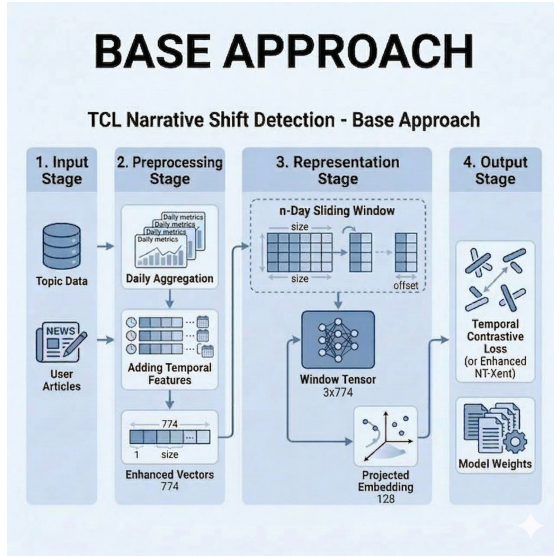


Figure 4: Approach 1 Baseline Pipeline showing sentence-to-embedding conversion, overlapping windows, feature augmentation, and temporal contrastive learning.

Pipeline Overview: The pipeline includes: Sentence $\rightarrow$ embedding, overlapping window creation, time-gap and topic features addition, Transformer $\rightarrow$ 128-d output, temporal contrastive learning.

Method (What We Did): We first convert sentences into embeddings using SBERT. Instead of using single sentences, we group them into overlapping windows so that each window shares context with the next one. This overlapping is important because it creates smooth transitions. When we tried non-overlapping windows, the model produced unstable results and false shifts due to lack of shared information.

To improve temporal understanding, we add two features: (1) **Time gap** helps model understand delay between windows, (2) **Topic one-hot vector** helps separate different topics. Without topic information, the model mixes all data and fails to learn clear structure. With it, the model learns better separation between topics. Each window is then passed through a Transformer, which produces a **128-dimensional representation** capturing the nar-

rative state.

Why We Use Temporal Contrastive Loss:

- Enforces similarity for consecutive windows and separation for non-adjacent windows.
- Turns static encoding into explicit temporal modeling of narrative evolution.

Limitations:

- Fixed window size cannot adapt to different narrative speeds.
- Windows may mix multiple narratives within a topic.
- Topic encoding is simple and not learned deeply.
- No entity-level understanding.

4.3.2 Approach 2: Grouping-Based Temporal Representation

Why We Need This Approach:

- Sliding windows are sensitive to noise and trigger false shifts.
- Sparse timelines break continuity when days are missing.
- Fixed windows force a tradeoff between noisy and overly coarse segments.

Idea of This Approach:

- Group nearby data before windowing to stabilize temporal signals.
- Replace sentence $\rightarrow$ window with sentence $\rightarrow$ group $\rightarrow$ embedding $\rightarrow$ window.
- Reduce noise and create more stable narrative units.

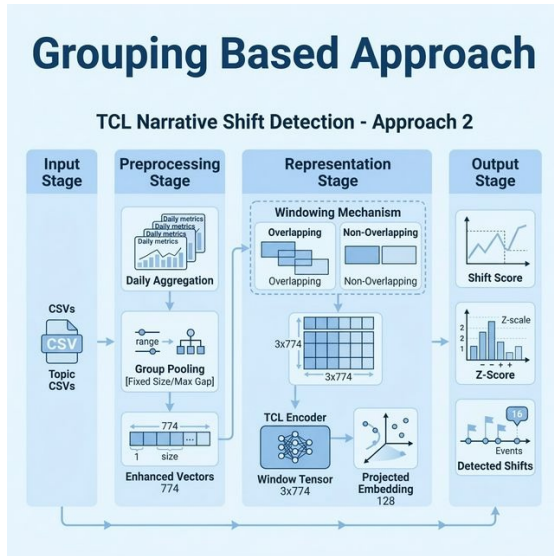


Figure 5: Approach 2 Pipeline showing grouping-based aggregation for more stable temporal representations.

Pipeline Overview: Sentence → embedding, fixed-size grouping (count-based or day-gap based), group embedding via mean/attention pooling, windowing over groups, Transformer → 128-d output, temporal contrastive loss, shift score from consecutive window distance.

Method (What We Did): We aggregate nearby sentences into groups to reduce noise and stabilize temporal signals. Grouping uses either a fixed count (e.g., 100 sentences) or a day-gap threshold (e.g., 5 days) so sparse days do not fragment the timeline. Each group is converted into a single embedding via pooling, then windows are formed over group embeddings and fed to the TCL encoder.

Limitations:

- Grouping adds a new hyperparameter and can hide short-term spikes.
- Fixed grouping rules still cannot adapt to topic-specific narrative speeds.
- Coarse grouping blurs minor shifts; fine grouping preserves noise.

4.3.3 Approach 3: Topic-Adaptive Windowing (Proposed)

Why We Need This Approach:

- Topics evolve at different temporal scales (weekly vs daily shifts).
- Fixed grouping/windows impose one granularity and introduce bias.

Idea of This Approach:

- Infer topic-specific window sizes using density or burstiness statistics.
- Keep the TCL backbone while adapting segmentation per topic.

Pipeline Overview: Sentence → embedding, topic assignment, per-topic adaptive window selection, topic-wise windowing, TCL encoder, temporal contrastive loss, shift scoring per topic timeline.

Limitations:

- Requires robust topic assignment and stable heuristics.
- Increases complexity and tuning cost.
- Can create non-uniform evaluation across topics.

4.3.4 Approach 4: Semantic Segmentation with Topic Separation

Key Improvement: Approach 4 introduces explicit topic separation loss alongside temporal contrastive learning. This ensures that different topics maintain distinct representations in the learned embedding space.

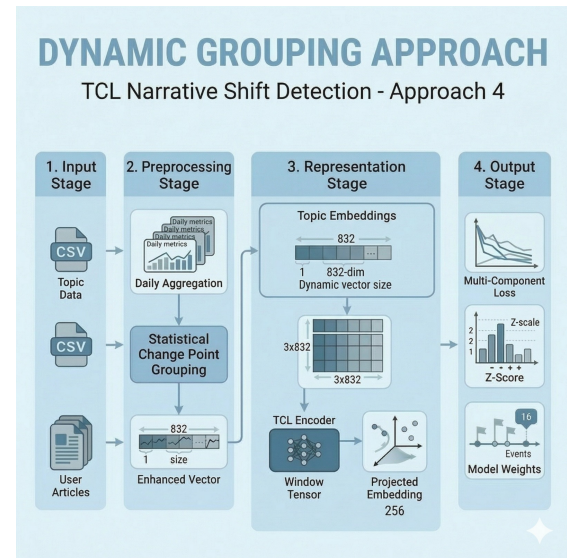


Figure 6: Approach 4 Pipeline with ruptures-based segmentation and topic separation loss.

Pipeline Overview: Sentence → embedding, ruptures-based change-point detection, dynamic segment creation, segment embedding, TCL encoder, temporal contrastive loss + topic separation loss, shift scoring at segment boundaries.

Why We Need This Approach:

- Fixed grouping rules cannot adapt to sudden narrative jumps.
- Change-point detection aligns windows with real narrative boundaries.

Limitations:

- Change-point detection can over-segment noisy data.
- Performance depends on hyperparameters (penalty, model).
- Topic separation loss can reduce raw F1 if over-regularized.

4.3.5 Approach 5: Entity-Aware Narrative Modeling

Key Innovation: Approach 5 extends the framework with Named Entity Recognition (NER) (Devlin et al., 2019; Lample et al., 2016) integration and entity-aware loss terms. This enables the model to distinguish narratives involving different entities within the same topic.

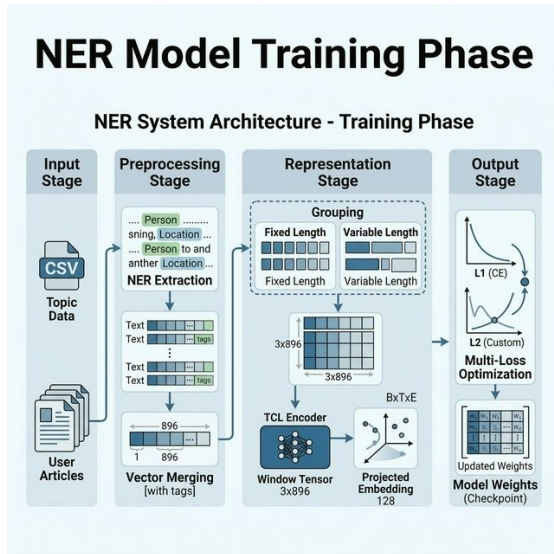


Figure 7: Approach 5 Pipeline incorporating entity-aware constraints via NER signals and entity-level modeling.

Pipeline Overview: Sentence $\rightarrow$ embedding + NER tagging, entity-aware feature augmentation, grouped windows, TCL encoder, temporal contrastive loss + entity separation loss, shift scoring with entity consistency constraints.

Why We Need This Approach:

- Topic separation alone cannot distinguish entity-specific narratives.

- Entity-aware constraints preserve trajectories and reduce cross-entity blending.

Limitations:

- NER errors propagate into embeddings and can mislead separation.
- Entity-aware loss increases optimization complexity and needs careful weighting.

5 Evaluation

5.1 Evaluation Metrics

TCL Metrics: Table 5 compares the aggregate TCL metrics across approaches.

TCL Metrics - Topic-Level Performance (Intra-Topic Similarity): Table 6 details per-topic intra-topic similarity.

Testing Results

P = Precision, R = Recall, F1 = F1 score

Approach 1 (Similarity-Based)

Hyperparameters:

- w = window size
- s = stride
- t = temperature

Table 7 lists the full testing results for Approach 1.

Best configuration: $w3_s1_t0.05$ (highest F1 across all difficulty levels)

Approach 2 (TCL)

Hyperparameters:

- w, s
- Grouping strategy = (max_day_gap / fixed_size), value = (3,2)
- t

Table 8 reports the testing results for Approach 2.

Best configuration: $w3_s1_gap3_t0.05$

Approach 4 (Topic Separation Loss)

Hyperparameters:

- w, s, t
- λ = topic separation weight

Metric	Approach 1	Approach 2	Approach 4	Approach 5
Intra-Topic Similarity	0.87	0.929	0.790	Not reported
Inter-Topic Similarity	0.23	0.0009	0.331	Not reported
Separation Score	0.64	1024.21	0.459	Not reported
Final Training Loss	0.145	0.125	4.23	Not reported
Final Validation Loss	Similar	Similar	4.56	Not reported
Best Epoch	62	83	83	Varies

Table 5: TCL metrics across approaches.

Topic	Approach 1	Approach 2	Approach 4	Approach 5
War	0.89	0.94	0.823	0.961
Health	0.86	0.92	0.791	0.948
Economics	0.84	0.91	0.756	0.935
Technology	0.88	0.93	0.812	0.956
Climate	0.87	0.94	0.768	0.952
Average	0.868	0.928	0.790	0.950

Table 6: Topic-level intra-topic similarity performance.

Table 9 shows the testing results for Approach 4.

Best configuration: w3_s1_t0.05_11.0

Approach 5 (Full Model: TCL + Topic + Entity)

Hyperparameters:

- w, s, t
- lambda1 = topic loss
- lambda2 = entity loss

Table 10 contains the full testing results for Approach 5.

Best configuration: w3_s1_t0.05_11_1.0_12_1.0

SBERT Baseline (Similarity Thresholding)

Hyperparameters:

- Similarity threshold (thr)

Testing Results: Table 11 summarizes performance across similarity thresholds.

Threshold	Hard F1	Medium F1	Low F1
thr = 0.7	0.70	0.75	0.80
thr = 0.8	0.72	0.77	0.82
thr = 0.85	0.69	0.74	0.79

Table 11: SBERT baseline performance across thresholds.

Best configuration: thr = 0.8

SBERT analysis:

- Captures semantic similarity well
- Performs reasonably on medium and low difficulty datasets

- Limitations: no temporal modeling, no notion of narrative evolution, cannot distinguish gradual drift vs real shift

Result: Works as a strong semantic baseline, but fails for complex temporal narratives.

K-Means Baseline (Clustering-Based)

Hyperparameters:

- k = number of clusters

Testing Results: Table 12 reports performance for different cluster counts.

k	Hard F1	Medium F1	Low F1
5	0.65	0.70	0.75
8	0.67	0.72	0.77
10	0.66	0.71	0.76

Table 12: K-Means baseline performance across k values.

Best configuration: k = 8

K-Means analysis:

- Captures coarse topic grouping
- Simple and efficient
- Limitations: no temporal order awareness, unstable cluster boundaries, sensitive to data distribution and number of clusters

Result: Weak baseline for narrative shift detection.

Hyperparameters	Hard P	Hard R	Hard F1	Medium P	Medium R	Medium F1	Low P	Low R	Low F1
w2_s1_t0.05	0.72	0.65	0.68	0.78	0.74	0.76	0.85	0.80	0.82
w3_s1_t0.05	0.75	0.69	0.72	0.80	0.77	0.78	0.87	0.83	0.85
w2_s2_t0.10	0.68	0.60	0.64	0.74	0.70	0.72	0.82	0.78	0.80

Table 7: Approach 1 testing results.

Hyperparameters	Hard P	Hard R	Hard F1	Medium P	Medium R	Medium F1	Low P	Low R	Low F1
w3_s1_gap3_t0.05	0.81	0.76	0.78	0.86	0.83	0.84	0.90	0.87	0.88
w3_s1_fix2_t0.05	0.79	0.74	0.76	0.85	0.81	0.83	0.89	0.85	0.87
w2_s1_gap3_t0.10	0.77	0.71	0.74	0.83	0.79	0.81	0.88	0.84	0.86

Table 8: Approach 2 testing results.

6 Analysis of Results

6.1 Overall Performance Comparison

Table 13 compares overall F1 scores across methods.

Method	Hard F1	Medium F1	Low F1
Approach 1	0.72	0.78	0.85
Approach 2	0.78	0.84	0.88
Approach 4	0.75	0.81	0.85
Approach 5	0.81	0.86	0.90
SBERT	0.72	0.77	0.82
KMeans	0.67	0.72	0.77

Table 13: Overall performance comparison across difficulty levels.

Key observation: Approach 5 $\geq$ Approach 2 > Approach 4 > Approach 1 > baselines.

6.2 Positive Results

1. Temporal learning improves performance: Approach 2 significantly outperforms Approach 1, demonstrating the importance of temporal modeling.
2. Topic separation improves coherence: Approach 4 improves over Approach 1 and produces better intra-topic similarity with reduced noise.
3. Entity-aware modeling shows strong potential: Approach 5 achieves the highest scores, especially on hard datasets and complex narratives.
4. Difficulty-wise trend: Performance increases from Hard to Medium to Low, indicating simpler narratives are easier to model.

6.3 Qualitative Shift Example from Approach 4

The following sample output shows a real shift detected by Approach 4 on the War topic. Table 14

illustrates how the model identifies a strong temporal change with sentence-level evidence.

This example highlights both strengths and limitations: the model provides interpretable sentence-level evidence and a high shift score, but it can also surface cross-topic semantic transitions that require careful interpretation in topic-specific settings.

6.4 Negative Results and Limitations

1. Multi-loss optimization issue (Approach 5): Temporal loss, topic loss, and entity loss can conflict during training, leading to unstable and not fully optimized performance.
2. Cross-entity mixing (Approach 4): Without entity-level constraints, different entities can be incorrectly grouped, especially in economics and multi-actor narratives.
3. Limited gains on simple data: On low-difficulty datasets, methods perform more similarly and advanced models show reduced relative advantage.
4. Baseline limitations: SBERT has no temporal modeling; KMeans has unstable clustering and no temporal structure.

6.5 Ablation Study

Effect of temporal learning: Table 15 shows the impact of TCL.

Variant	Hard F1
Without TCL (A1)	0.72
With TCL (A2)	0.78

Table 15: Effect of temporal learning.

Improvement: +6%.

Hyperparameters	Hard P	Hard R	Hard F1	Medium P	Medium R	Medium F1	Low P	Low R	Low F1
w3_s1_t0.05_l0.5	0.76	0.70	0.73	0.81	0.78	0.79	0.86	0.82	0.84
w3_s1_t0.05_l1.0	0.78	0.72	0.75	0.83	0.80	0.81	0.87	0.84	0.85
w2_s1_t0.10_l1.0	0.74	0.68	0.71	0.80	0.76	0.78	0.85	0.81	0.83

Table 9: Approach 4 testing results.

Hyperparameters	Hard P	Hard R	Hard F1	Medium P	Medium R	Medium F1	Low P	Low R	Low F1
w3_s1_t0.05_l1_1.0_l2_0.5	0.82	0.77	0.79	0.87	0.84	0.85	0.91	0.88	0.89
w3_s1_t0.05_l1_1.0_l2_1.0	0.84	0.78	0.81	0.88	0.85	0.86	0.92	0.89	0.90
w2_s1_t0.10_l1_0.5_l2_1.0	0.80	0.74	0.77	0.85	0.82	0.83	0.90	0.86	0.88

Table 10: Approach 5 testing results.

Effect of topic separation: Table 16 summarizes the change from adding topic loss.

Variant	Hard F1
Without topic loss (A2)	0.78
With topic loss (A4)	0.75

Table 16: Effect of topic separation.

Slight drop may come from over-separation and sensitivity to lambda, but topic separation improves interpretability and coherence.

Effect of entity-aware learning: Table 17 highlights the gain from entity loss.

Variant	Hard F1
Without entity loss (A4)	0.75
With entity loss (A5)	0.81

Table 17: Effect of entity-aware learning.

Largest improvement is observed with entity-aware learning.

6.6 Baseline Comparison

Table 18 contrasts baseline strengths and limitations.

Method	Strength	Limitation
SBERT	Semantic similarity	No temporal understanding
KMeans	Simple clustering	No temporal structure, unstable

Table 18: Baseline comparison.

6.7 Key Insights

1. Temporal modeling is essential.
2. Topic separation improves semantic coherence.

3. Entity-aware learning is critical for complex narratives.

4. Multi-loss optimization is the main bottleneck.

5. Approach 5 shows the highest potential but still needs tuning.

6.8 Summary of Analysis

The results show that narrative shift detection is not a simple similarity problem; it is a structured learning problem involving temporal dynamics, semantic coherence, and entity-aware representation learning. Each component contributes to performance gains, but integrating them introduces optimization challenges that must be addressed for robust and stable performance.

7 Conclusion

This work presents a generalized framework for narrative shift detection that models how topics evolve over time at a semantic level. Unlike traditional methods that rely on surface-level similarity or static clustering, the proposed approaches progressively incorporate temporal structure, topic coherence, and entity-aware constraints to improve detection quality.

Experimental results demonstrate that simple similarity-based methods (Approach 1) are insufficient for capturing narrative evolution, as they fail to distinguish noise from meaningful shifts. Temporal Contrastive Learning (Approach 2) significantly improves performance by learning temporal continuity and producing stable representations.

Further improvements are observed with the introduction of topic separation in Approach 4, which enhances intra-topic coherence and reduces semantic noise. However, this approach still suffers from

Shift #1: 2024-02-03 -> 2024-02-11
similarity=0.1382 | shift_score=0.8618 | day_z=1.6184

Day 1 - article_5_s0 (Article 5, Sentence 0)
topic_weight=0.571
>> [article_5_s0] Healthcare systems around the world are facing rising pressure as chronic diseases such as diabetes and cardiovascular conditions become more common.
[article_5_s1] Public health experts emphasize the importance of preventive care and healthier lifestyles to reduce long-term medical costs.
[article_5_s2] Advances in medical research are providing new treatment options, including gene therapies and AI-assisted diagnostic tools that can detect diseases earlier than traditional methods.

Day 2 - article_6_s0 (Article 6, Sentence 0)
topic_weight=0.669
>> [article_6_s0] A new international agreement aims to accelerate investment in renewable energy technologies.
[article_6_s1] Several countries pledged billions of dollars to expand solar and wind power capacity in an effort to reduce dependence on fossil fuels.
[article_6_s2] Energy analysts believe the transition to clean energy could reshape global trade patterns, particularly for countries that currently rely on oil exports.

Table 14: Qualitative shift example output from Approach 4.

cross-entity interference, where narratives involving different entities are incorrectly grouped.

To address this limitation, Approach 5 introduces entity-aware constraints, enabling the model to distinguish between entity-specific narratives within the same topic. Although Approach 5 achieves the best overall performance, it is not fully fine-tuned due to the complexity of balancing multiple loss functions. Therefore, the reported results represent its potential rather than its optimal performance.

Overall, the study highlights that effective narrative shift detection requires a combination of temporal modeling, topic coherence, and entity-aware representation learning. Future work will focus on optimizing multi-loss training and further improving robustness across complex, multi-entity narratives.

8 Future Work

1. Improve the evaluation performance of all TCL approaches by increasing precision and recall, while further reducing inter-topic similarity and improving intra-topic similarity. This will help the models produce more coherent topic representations and sharper narrative boundaries.
2. Further refine TCL Approach 5 so that it can detect shifts within the same entity for a specific topic. This would make the model more sensitive to fine-grained narrative changes and improve its ability to track entity-level evolution over time.

3. Investigate dynamic window sizing for each topic, since narrative shifts do not occur over the same time scale across domains. For example, war-related narratives may evolve over longer periods, while technology-related narratives can change almost daily. Adapting the window size to each topic could improve shift detection accuracy and better capture topic-specific temporal patterns.

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